Sample-based index evaluation

Reviewer note (COL868)

PostgreSQL 14, TPC-DS SF=1, 99 queries, 33 configurations, 5% samples, one seed.
5–6 October 2026.

1 Question

Optimizer cost is an unreliable ranking of indexes (Wu et al., SIGMOD 2025). Timing every candidate on the full database is too expensive. **Does a 5% sample recover the same execution-time ranking?** Closest prior work, SQL:Trek (PVLDB 2025), uses one 5% Bernoulli recipe and does not report agreement with a full-database ranking of the same configurations.

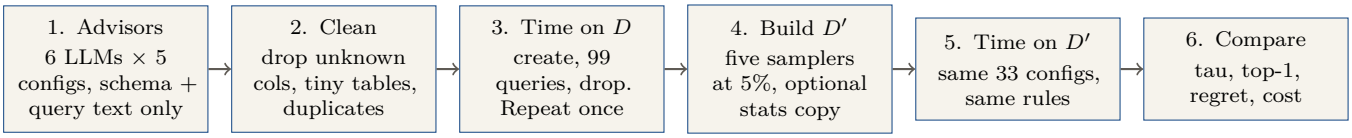
Supported here: on this SF=1 run, cheap random samples pick a configuration within 1.4% of the full-database winner, at 25–90× lower eval cost, and rank no worse than the full database ranks itself (tau ≈ 0.45). Join-aware sampling and copying full statistics did not help.

Not supported: that this holds at 100 GB, on SQL Server, or that any sampler is strictly better than another.

2 What was run

Assignment	This run	Why
SQL Server, 100 GB TPC-DS	PostgreSQL 14, SF=1 (∼1 GB, in memory)	SF=10 queries ran for hours on a 16 GB laptop.
Samples at 1%, 5%, 10%	5%, one seed	Time.
Best index per query	Whole-workload configuration (1–8 indexes)	Per-query winners flipped between identical full-DB runs.
Cost and time	Ranked by measured time (A). Score B = build + time.	The question is execution time.
Random / stratified / stats / workload-aware	Bernoulli, page, universe, FK, stratified; stats copy on two. No workload-aware.	

3 How it is implemented



Candidates. Isolated agents see schema, existing indexes, and the 99 query texts. They do not see data or previous timings. One Gemini run opened earlier result files and was thrown out. After cleaning: 33 configurations. Baselines: no extra index, HypoPG greedy top-5 and top-8 (ranked by estimated cost, then timed for real).

One configuration at a time. `CREATE INDEX` (timed), run all 99 queries, `DROP`. Leftover indexes from a killed run are dropped first so they cannot leak into the next config.

Timing a query. Autovacuum off. `ANALYZE` once per database, then frozen (a stats fingerprint is stored with every result row). Discard one warm-up. Then two timed runs if that warm-up was ≤ 2 s, else one. Median. Cap 20 s, counted as 20 s on both D and D'.

Scores. A = sum of 99 query times (primary). B = index-build time + A. Lower is better. A timeout or error is 20 s.

Comparison. Kendall tau-b with a 95% bootstrap over the 33 configs. Top-1 is exact. Regret is D's time of D''s pick divided by D's best (1.014 = 1.4% slower). Cost reduction is (build + workload over all configs on D) / (the same on D').

3.1 How a sample is built

Target size per table T follows SQL:Trek: keep

$$k_T = \min(\max(p\,n_T,\ 20{,}000),\ 10^6 \cdot (p/0.05),\ n_T)$$

at $p = 0.05$. Small tables are oversampled; huge tables are capped.

Sampler	Rule
Bernoulli	Each row of each table kept independently with probability k_T/n_T. Shrinks facts and dimensions (∼230 MB).
Page	Same size, `TABLESAMPLE SYSTEM` (whole disk pages). Cheapest to build.
Universe (item)	Keep a fact row iff hash(`item_sk`) mod $10^4 < 100p$. Same hash in every fact that joins on item. Dimension tables kept in full (∼530 MB).
FK closure	Bernoulli on the three sales facts. Keep matching returns. Thin inventory among sampled items. Dimensions in full.
Stratified	On sales, keep at least 5 rows per sold-date, then Bernoulli. Other facts as Bernoulli. Dimensions in full.
Stats transplant	Same physical rows. Copy full-DB column statistics and set `reltuples`/`relpages` to the full values. Do not `ANALYZE` after. Indexes on the sample stay small.

Independent 5% on two join partners keeps about 0.25% of matches. That is why universe and FK exist. They pay for it by leaving dimensions whole, so they stay large.

4 Results

No extra index: **190.7 s**. Best (Grok, small dimension filters): **172.9 s**. Spread 17.7 s.



Figure 1: No-index total 190.7 s. Seven queries hit the 20 s cap under every configuration, including the best, so 140 s of every total cannot move. Indexes only rearrange the other ∼50 s.

A second full run the next morning: same winner and second place, **tau 0.45** [0.18, 0.67], every config ∼5.3 s slower. That 0.45 is the noise ceiling.

Sample	tau	95% CI	Top-1	Regret	Top-3	Cheaper	Pick
Bernoulli	0.51	0.26–0.72	no	1.014	yes	43× (25×)	Opus light
Page	0.46	0.20–0.69	no	1.014	yes	90× (33×)	Opus light
Universe	0.45	0.27–0.63	no	1.014	yes	2.6×	Opus light
FK closure	0.42	0.17–0.62	no	1.014	yes	3.5×	Opus light
Stratified	0.28	−0.01–0.56	yes	1.000	yes	3.4×	Grok small
Bernoulli, full stats	0.24	0.01–0.50	no	1.014	no	30×	Opus light
Universe, full stats	0.30	0.03–0.52	no	1.078	yes	2.4×	Composer mixed

Cost in parentheses includes building the sample. Grok small is the true best (172.9 s). Opus light is 175.4 s (1.4% slower). Native samples always put those two in the top 2. Score B does not change picks except stratified, which then also misses. Copying full statistics made both rankings worse. Intervals overlap; one seed.

The two winners mainly cut Q6 (20 s cap → ∼10 s) and Q14 (7.0 → ∼3.7 s), both via date-dimension filter indexes. Inferred from per-query times, not an ablation.

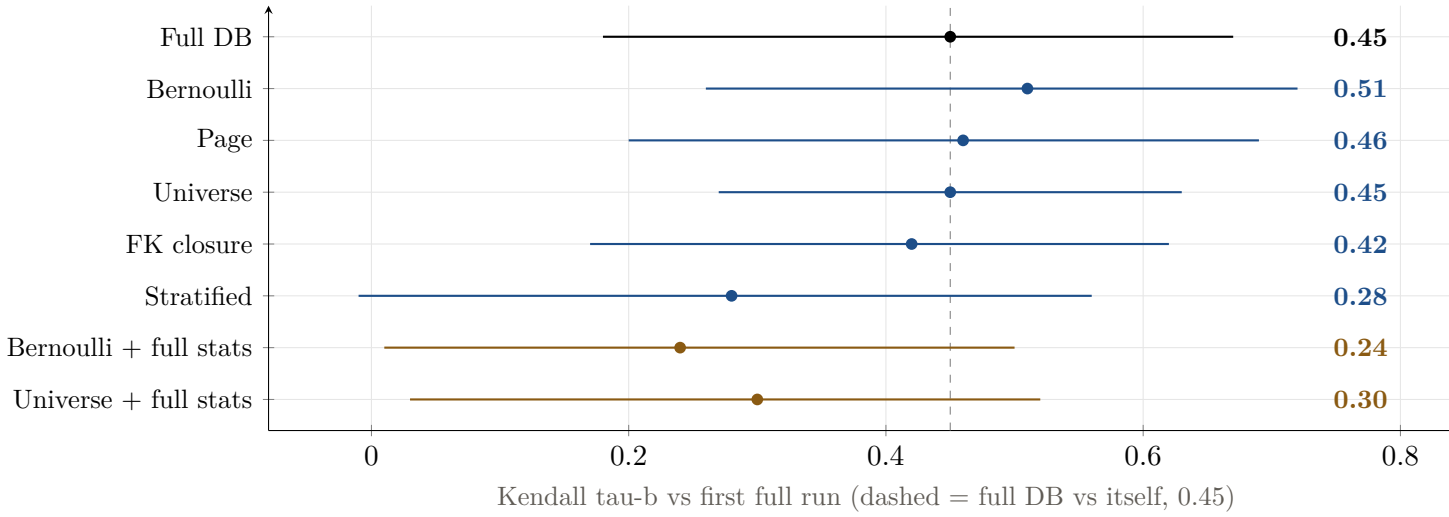


Figure 2: 95% bootstrap CI over 33 configs. Blue: native 5%. Brown: same rows, full-DB statistics copied on.

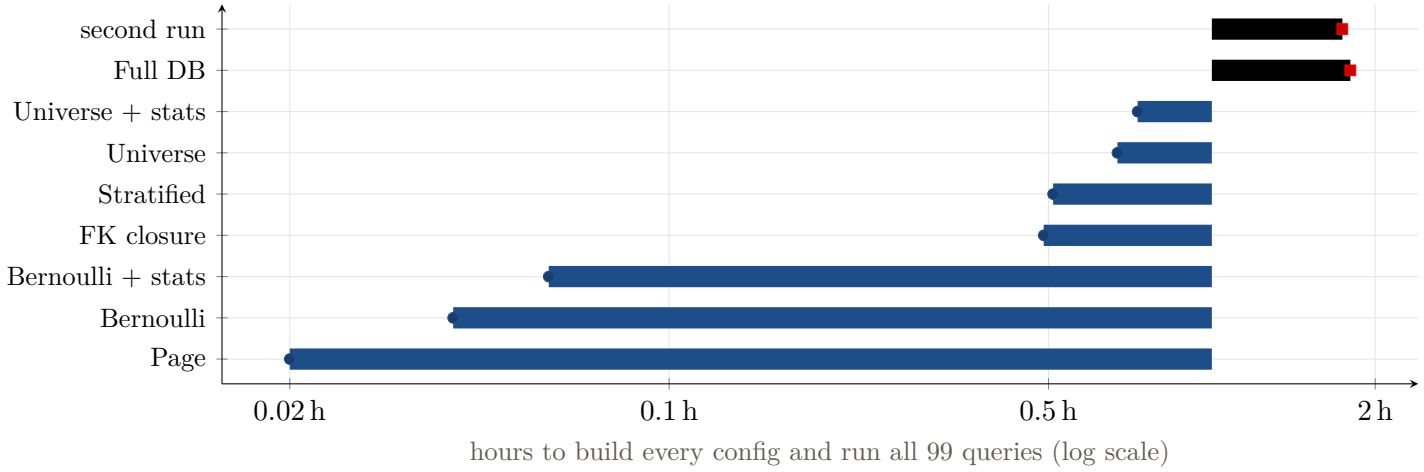


Figure 3: Join-aware samples keep dimension tables whole, so they only save ∼3×. Bernoulli and page shrink every table.

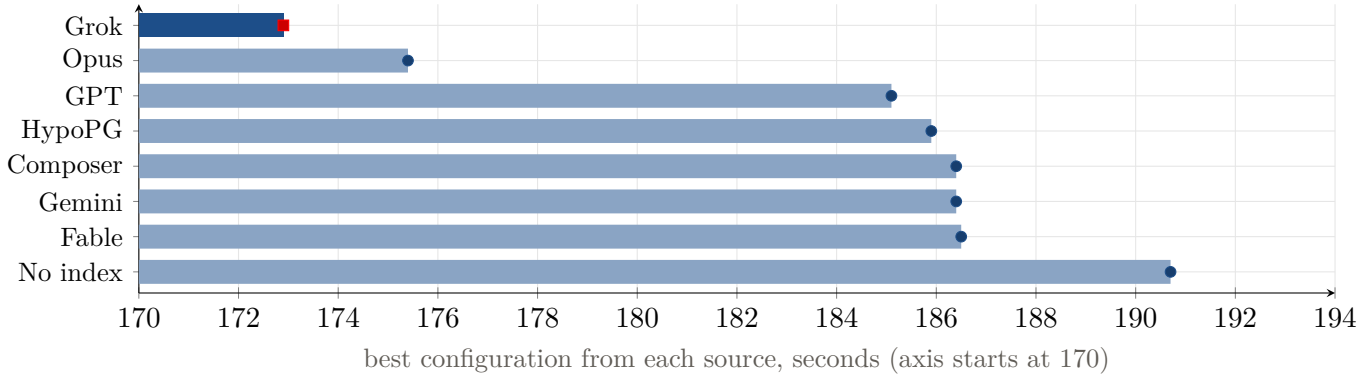


Figure 4: HypoPG's greedy set, ranked by optimizer cost then timed, lands in the middle. Empty is the slowest of the 33. Only Grok and Opus moved the needle much.

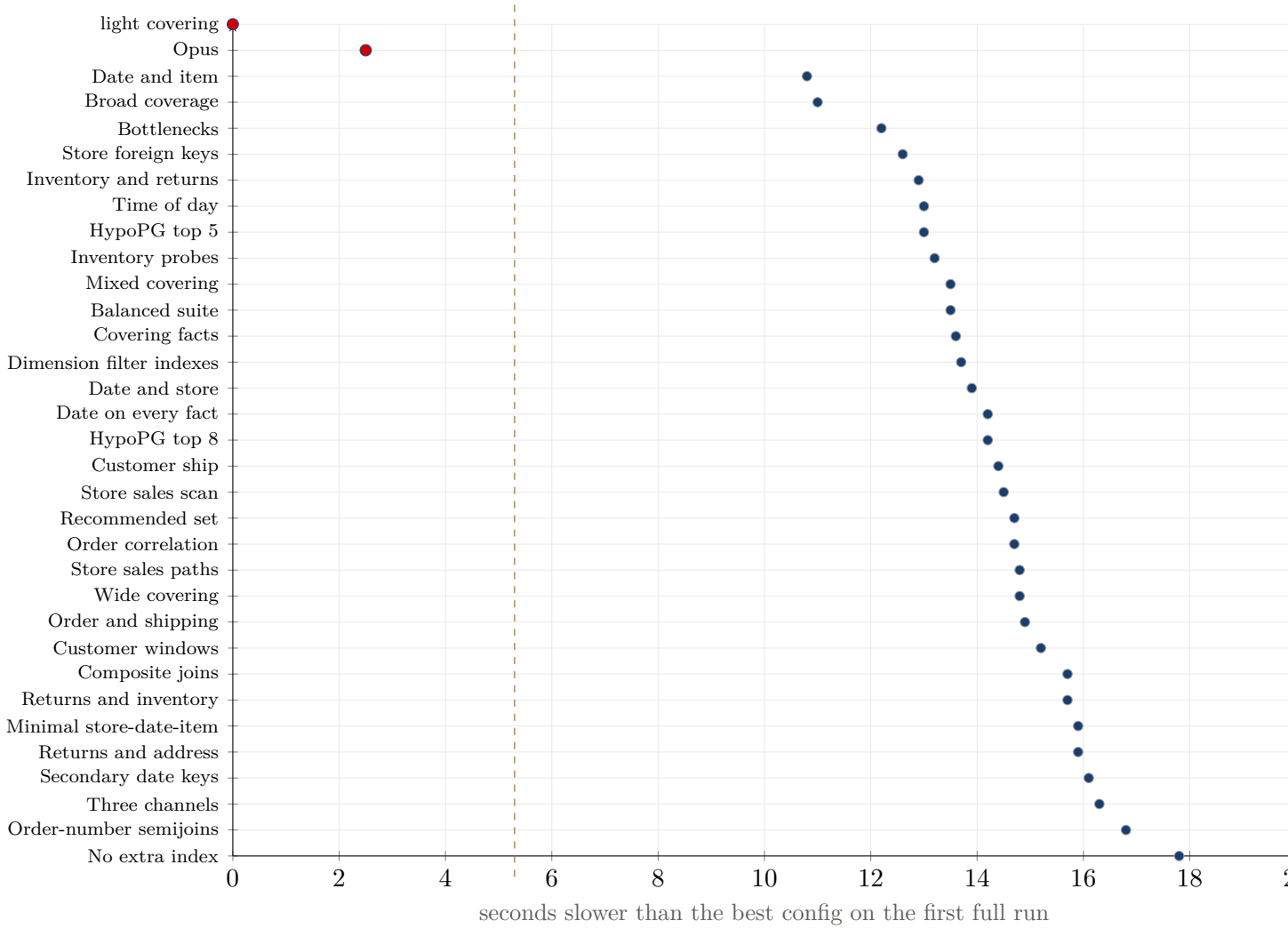


Figure 5: Dashed line: 5.3 s median overnight shift. Winner and second are 2.4 s apart, inside that shift. 31 of 33 configs sit within 4%. That is why tau against a repeat of the full database is only 0.45.

Expectation	Outcome
Sample finds the execution-time best.	Exact top-1 on 1 of 5 native samples. Others 1.4% off; true best always in top 3.
Join-aware beats random.	No. Same quality, $\sim$3$\times$ cheaper vs 40–90$\times$.
Full statistics improve fidelity.	No, on the two samples tried.
Samples rank like the full DB.	To within noise (tau 0.42–0.51 vs ceiling 0.45).

5 Caveats

- One seed, one rate, $n = 33$ near-ties. Sampler differences of 0.05 in tau are inside the CIs.
- 20 s cap hides any index that would have helped the seven always-capped queries, including Q6 under no-index.
- SF=1 is cache-resident. Assignment was 100 GB SQL Server.
- Stats transplant copies table stats only; indexes stay small. Negative result is for this implementation.
- Candidate pool is clustered. Empty is the only clearly bad config.
- Measurement choices (cap, score A, SF=1) were informed by an earlier noisy per-query pilot, not blind.

6 What to push on

1. Is 1.4% regret a success when best-to-worst is only 9% and overnight drift is 5.3 s?
2. Score A or B? They disagree on stratified top-1.
3. Is tau-vs-self 0.45 a fair ceiling from one repeat?
4. Next measurement that would settle the question: SF=10 on a 64–128 GB machine, two full rankings, then Bernoulli + page at 1/5/10% plus one join-consistent sampler, extra seed. More SF=1 cells will not.